POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1) Enlist the greatest breakthrough in microprocessor so that modern processors are available for personal computer. 5
2) With reference of bus advancement differentiate between Harvard Architecture and Von Neumann architecture. 5
3) Explain with example the addressing mode of 8085 microprocessor. 5
4) Define instruction set and formats, opcodes, mnemonics and operands with examples. 8
5) Write an 8085 program for the following type of addition 7

$$1^2 + 2^2 + 3^2 + \cdots + 9^2$$

a) Define T-state. Draw the timing diagrams for the instruction OUT 0FH. 8
b) Write a program to take input from 4- switches connected to PC3-PC0 and display the status of the switches to 4 LEDs connected to PC7 – PC4 of 8255 PPI. 7
a) What is memory interfacing? Interface two PROM and one RAM with 8KB and 4KB memory respectively with 8085. 8
b) Explain the Programmable Interrupt Controller (8259 A) with a suitable block diagram. 7
a) Do you thing micro assembler is essential for microprocessor to execute the program? If yes, explain with the difference between 7
   i. END and ENDP
   ii. DB and DW
   iii. PROC and EXTRN
   iv. PUBLIC and GLOBAL
b) Write an 8086 program to find square root of a given number. Given that number is a perfect square of two digits. 8
6. a) Explain with example the flags of 8086 microprocessor.
   b) Write the difference between serial and parallel communication. Explain the RS232C standard for serial communication.
7. Write short notes on: (Any two)
   a) 8255 PPI
   b) 80386 microprocessor and its type
   c) Procedures and Macros

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